

River basins as groundwater exporters and importers: Implications for water cycle and climate modeling

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[1] The groundwater reservoir and its interaction with surface water facilitate lateral transport of continental water and energy. Current climate models do not account for long-distance groundwater flow between model cells but route the atmospheric surplus (precipitation (P) minus evapotranspiration (ET)) directly to stream discharge within a model grid cell. We ask how much water exits a river basin without ever passing through its surface outlet? What are the climatologic and geologic factors influencing this flux? To answer these questions, a separation of groundwater flow from river flow is necessary. We use the ratio of stream discharge (Qr) to basin recharge ($R = P - ET$) for this purpose; where $Qr:R < 1$, a basin is considered a groundwater exporter; and where $Qr:R > 1$, a basin is considered a groundwater importer. Here Qr is obtained from 39 years of U.S. Geological Survey Hydro-Climatic Data Network observed stream discharge from 1555 basins across the continental United States, and $R (P - ET)$ is derived from 50 years of hydrologic simulation by the Variable Infiltration Capacity model. It was found that the Qr:R ratio deviates significantly from 1 across the continent. Detailed investigations of individual basins suggest that the deviations are primarily a function of geology, while climate and basin scale influence the magnitude of these deviations. Further, a marked incongruity between the surface and groundwater flow directions is apparent, suggesting that surface drainage is only partially indicative of subsurface flow regimes. The apparent significance of this long-distance groundwater flow component reinforces the need for inclusion of the groundwater reservoir in current water cycle and climate modeling efforts.

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1. Introduction

[2] The terrestrial water cycle is an inherent component of Earth's climate system and our ability to quantify its fluxes has direct relevance to improving water cycle and climate models. In this paper, we discuss one particular hydrologic flux: regional groundwater flow that exports water from one river basin and imports it to another, and its significance and implications to river basin water budget and terrestrial water cycle modeling.

[3] Figure 1a is a highly simplified illustration of land hydrology. When rain falls on the continent, it takes multiple pathways to return to the atmosphere or the ocean: (1) direct evaporation of intercepted rain from vegetation, (2) surface runoff into local rivers, lakes, and wetlands, which either evaporates or follows the surface drainage network into the ocean, (3) evapotranspiration (ET) of soil pore water filled during previous rain events, (4) upward capillary flux from the water table to root zone to supply ET in dry periods, (5) shallow groundwater flow and discharge

to local wetlands and streams, (6) deeper and longer-distance groundwater flow and discharge to wetlands and rivers further downstream, and (7) direct submarine groundwater discharge into the coastal ocean. While pathways 1–3 are associated with land surface and vadose zone processes, pathways 4–7 are entirely facilitated by the groundwater reservoir.

[4] It is recognized that pathways 1–3, particularly ET flux, have important implications to land-atmosphere energy partition and boundary layer dynamics, and that the key control of ET is root zone and near-surface soil moisture [e.g., Manabe, 1969; Avissar and Pielke, 1989; Entekhabi *et al.*, 1992; Eltahir, 1998; Hong and Kalnay, 2000; Small, 2001; Pal and Eltahir, 2002; D'Odorico and Porporato, 2004; Koster *et al.*, 2004; Betts, 2004]. This recognition has motivated generations of land surface schemes in state-of-art climate models, with detailed process descriptions and parameterizations of ET and runoff fluxes.

[5] However, pathways 4–7, which are associated with the groundwater reservoir, have not been systematically evaluated in the context of climate modeling until recently. Figure 1b depicts the continental reservoirs (within the red oval) and the fluxes (red arrows) included in a typical climate model. Normally the groundwater reservoir is not

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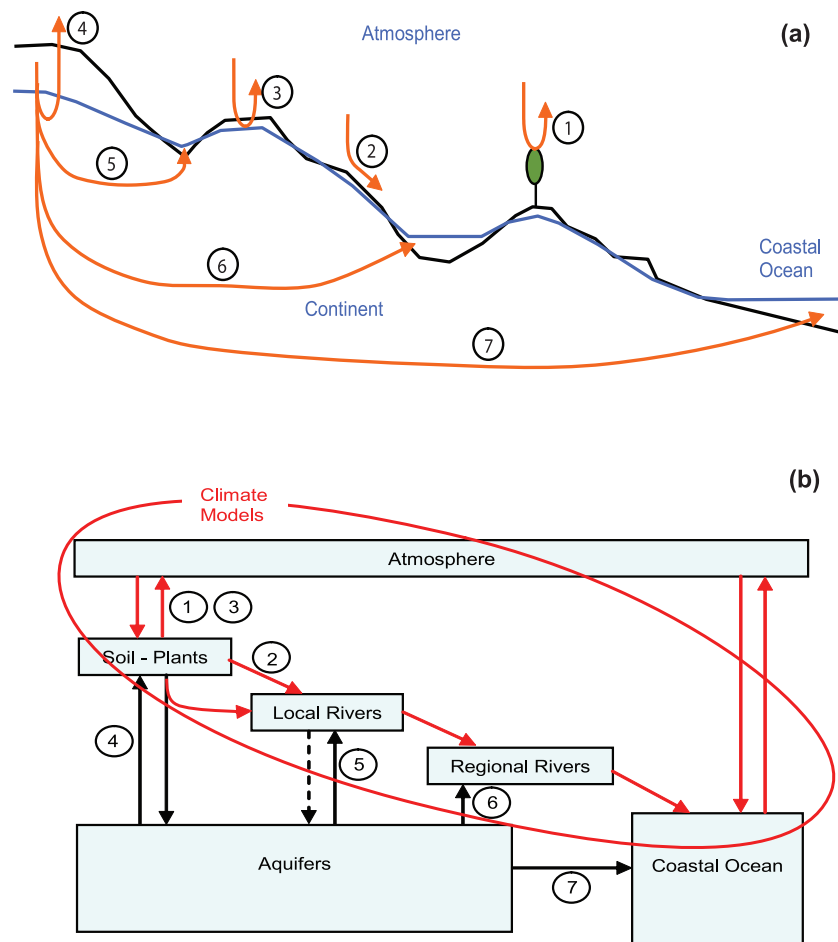


Figure 1. (a) A simplified schematic of hydrologic pathways, where pathways 4–7 are associated with the groundwater reservoir, and (b) reservoirs (within the red oval) and fluxes (red arrows) represented in current climate models.

included, inevitably excluding the pathways associated with it. A standard approach is to prescribe free gravity drainage at the base of the model soil column, which is then placed directly into the rivers within a model grid cell and routed through the river network into the ocean. This bypasses the deeper storage and flow paths operating at larger spatial and longer temporal scales. While this approach is adequate for short-term weather forecasts, it may not be appropriate for climate studies where slow processes are important. The questions are as follows: how important are pathways 4–7? To what degrees do they influence the water and energy budget of near-surface reservoirs, namely, the soil-vegetation and the surface water (wetlands, lakes, and rivers) which are more direct participants in the climate system?

[6] Pathway 4, the groundwater and the soil moisture link, has been the subject of several recent investigations of land-atmosphere coupling [e.g., Salvucci and Entekhabi, 1994, 1995; Gutowski *et al.*, 2002; Liang *et al.*, 2003; Maxwell and Miller, 2005; Yeh and Eltahir, 2005; Yu *et al.*, 2006; Niu *et al.*, 2007; Fan *et al.*, 2007; Miguez-Macho *et al.*, 2007, 2008]. It was found that where the water table is shallow it directly influences soil moisture and ET flux. Because groundwater operates at larger spatial and longer temporal scales, this link affects not only the amount, but

also the spatial-temporal variability in the overlying soil moisture fields [Miguez-Macho *et al.*, 2007, 2008].

[7] Pathway 5, the dynamic interaction between shallow groundwater and local stream flow, has long been studied by hydrologists seeking to understand runoff mechanisms at hillslope to watershed scales. The subject is elegantly discussed in many past and recent papers [e.g., Hewlett and Hibbert, 1963; Dunne and Black, 1970a, 1970b; Tanaka *et al.*, 1988; de Vries, 1994, 1995; Eltahir and Yeh, 1999; Marani *et al.*, 2001]. They provided the observational and theoretical basis for a mechanistic understanding of this coupling at river basin scales. A key mechanism is the growth and retreat of the river network in response to water table rise and fall; accelerating drainage when the water table rises, and shutting down drainage when the water table falls. This negative feedback stabilizes the water table depth and leads to an asymmetric water table response rate during the rise and fall of the water table [Eltahir and Yeh, 1999]. Attempts to incorporate this mechanism into climate models significantly improved river flow simulations and reduced the number of tunable parameters [e.g., Yeh and Eltahir, 2005; Miguez-Macho *et al.*, 2007].

[8] The focus of this paper is pathway 6, the deeper and longer-distance groundwater flow that transports water from



Figure 2. Map of the High Plains Aquifer with water table contours and general direction of groundwater flow [from *U.S. Geological Survey*, 1990–1999, available at http://capp.water.usgs.gov/gwa/ch_a/A-text2.html].

one river basin to another, and feeds the wetlands and rivers far downstream from its source (Figure 1a). In current climate models, this pathway is short-circuited; soil drainage is placed directly into local rivers within a model grid cell, regardless of grid size and routed to the ocean through the river network (Figure 1b). If deeper, long-distance groundwater flow is a significant part of a river basin's water budget, then our current approach may overestimate river flow in first-order streams in upper reaches and underestimate it in first-order streams of lower reaches; hence groundwater transport has several implications for climate simulations. First, the above effect introduces errors in the modeled river flow, and if the latter is forced to match

observed river flow by tuning ET parameters in order to close the water budget, the errors in modeled river flow may propagate into modeled ET. Second, via long-distance groundwater flow, atmospheric surplus or deficit (precipitation (P) – ET) is redistributed across the landscape, giving rise to a stronger spatial heterogeneity and organization in water table depth and soil moisture fields [Fan *et al.*, 2007; Miguez-Macho *et al.*, 2007]. It is well known that spatial organization in soil moisture can lead to horizontal gradients in atmospheric states, influencing circulation and vapor convergence [e.g., Avissar and Pielke, 1989; Small, 2001; Pal and Eltahir, 2002; Georgescu *et al.*, 2003; Kanamitsu and Mo, 2003]. Third, long-distance groundwater convergence

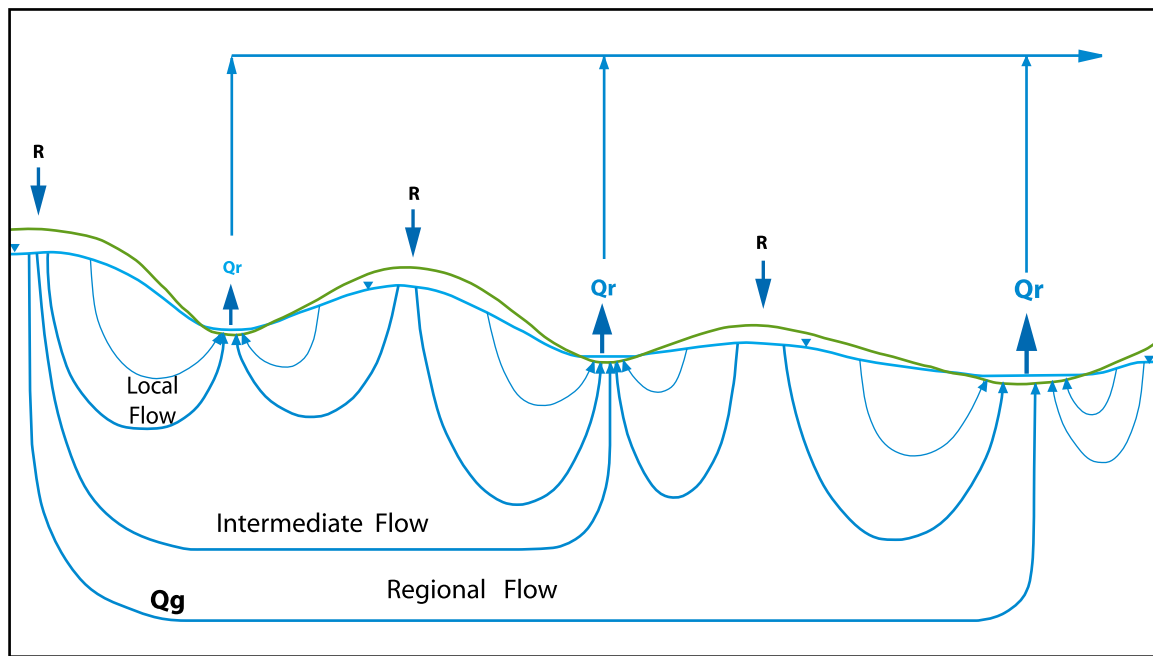


Figure 3. A schematic illustration of *Toth's* [1963] conceptualization of multiple-scale groundwater flow, where the local flow is nested within the intermediate flow which in turn is nested in the even larger regional flow system.

sustains lakes, wetlands and oases in arid and semiarid regions and coastal zones, which may increase ET over the discharge area, potentially enhancing local precipitation. Using a regional climate model, *Anyah et al.* [2008] shows that moist valley soils in Arizona, sustained by groundwater convergence from the snow-fed mountains, can significantly enhance ET in this water-limited environment, leading to a 50% increase in simulated convective rainfall during the 1997 North American Monsoon season. Fourth, via long-distance groundwater flow, climate anomalies at high-elevation recharge areas are transported to low-elevation discharge areas. This process results in the long-term persistence of soil water regimes in discharge areas, reflecting past climate conditions, often from far distances. A study by *Bierkens and van den Hurk* [2007] demonstrates that long-distance groundwater convergence can be a mechanism for soil water persistence over regions of groundwater discharge, locking precipitation into prolonged anomalies through positive soil moisture-rainfall feedbacks. Thus, our current understanding points to several potential links between long-distance groundwater flow and continental climate dynamics.

[9] There are numerous documented cases of long-distance groundwater flow across large regional aquifers. A noted example is the High Plains Aquifer in North America (Figure 2) where groundwater, sourced in the Rockies to the west, flows eastward in the blanket-sand-gravel aquifer, feeding successively larger streams to the east [*Gutentag et al.*, 1984; *Luckey et al.*, 1986; *Weeks et al.*, 1988]. This regional flow is particularly important during the more frequent dry periods when water table is below local relief, and flow is almost exclusively to major channels far to the east [*Gutentag et al.*, 1984]. Significant interbasin groundwater input to lowland streams is also observed in humid Costa Rica [*Genereux et al.*, 2002,

2005; *Genereux and Jordan*, 2006] and the Atlantic Coastal Plain [*Modica et al.*, 1997, 1998] based on chemical tracers. *Winter et al.* [2003] synthesized a large body of field evidence on the role of regional groundwater flow in sustaining lakes and wetlands located at the lower end of the groundwater flow system under a wide range of hydro-geologic conditions.

[10] In this paper, we present a first-order, quantitative assessment of the magnitude of this long-distance flow pathway (pathway 6) based on a simple water balance analysis. We define atmospheric surplus, or basin recharge ($R = P - ET$) as precipitation minus evapotranspiration ($R = P - ET$). We will use historically observed river flow and model-simulated land surface water-energy balance, over a 50-year period from the contiguous United States. We ask the following questions: for a given river basin, how much of the atmospheric surplus leaves the basin through the sub-surface pathways without ever entering the surface drainage network (e.g., 5% versus 50%), and under what climatic and geologic conditions is this flux a significant component of a river basin's water budget?

2. Theoretical Framework

2.1. Toth's Theory of Multiple-Scale Groundwater Flow

[11] The classic study of *Toth* [1963] is widely regarded as the theoretical framework for studies of groundwater flow [e.g., *Freeze and Cherry*, 1979] which embodies the following key concepts:

[12] 1. Groundwater flow can occur at multiple scales (Figure 3), with local flow nested inside larger, intermediate systems, which in turn are nested inside even larger regional flow systems.

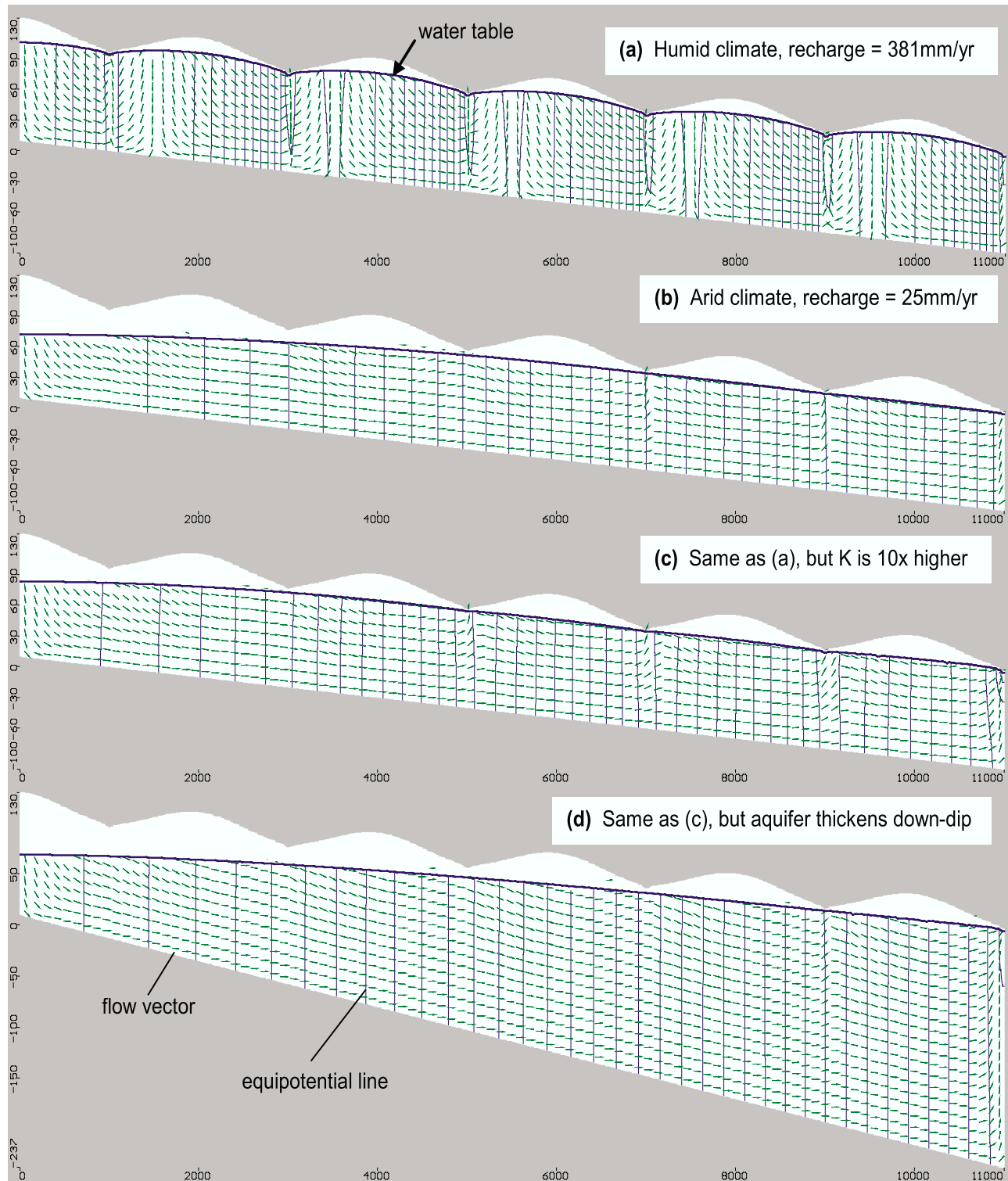


Figure 4. A conceptual model simulation of groundwater flow showing the effect of climate and geology. (a) In a humid climate with high recharge, the water table reflects local topography, and local drainage is active with gaining streams, and the regional flow system is diminished. (b) In an arid climate with low recharge, the water table falls below the upper valleys, streams are primarily losing, and groundwater discharges at regional lows. (c) A higher hydraulic conductivity leads to lower water table below the upper valleys and more significant regional flow. (d) A thickening aquifer in the down-dip direction further enhances regional flow.

[13] 2. Local flow, shallow and across short distances, has short residence times and is affected by short-term climatic fluctuations; while regional flow, deeper and across longer distances, has long residence times and is isolated from short-term climate variability.

[14] 3. The relative importance of each flow regime depends on several factors: large local relief enhances local flow due to deeply incised valleys acting as local drains, while large regional relief favors stronger regional flow due to enhanced hydraulic gradient at larger scales; shallow aquifers tend to force groundwater discharge to local rivers, while deep aquifers have more room to develop significant regional flow.

[15] Although not stated explicitly by *Toth* [1963], the following can be inferred:

[16] 4. Through intermediate and regional flow in strictly nested basins, higher elevation basins are groundwater exporters, and lower elevation basins are importers.

[17] 5. In each of the flow systems, there exists a hinge line, defining the area above as a net exporter and the area below as a net importer. In Figure 3, the hinge line of the local flow system lies between the ridge and adjacent valley. In the regional flow system, the hinge line lies between the uppermost ridge and the lowermost valley, but likely in the lower half of the basin.

[18] 6. The relative importance of local versus regional flow depends on basin size; small basins tend to be exporters or importers, and large basins tend to be self-contained. Over a continental-scale basin such as the Mississippi River Basin that has negligible direct submarine groundwater discharge, net exporters and importers tend to balance, and all recharge over the basin exits the outlet of the largest basin as river flow. This is achieved by default in *Toth's* [1963] study, which is based on numerical models with no-flow boundaries at both ends of the model cross section (Figure 3).

2.2. Extensions of Toth's Theory

[19] *Toth's* [1963] theory is largely based on a set of numerical simulations in a given climate with homogeneous geology. In applying the theory to a natural system, one encounters additional factors that also influence the flow regime. Below we extend *Toth's* study to include three additional observations based on conceptual model simulations, following *Toth's* methodology.

[20] 1. The climate of a basin must affect the water table depth and hence the flow regime. In the model simulation shown in Figure 4a, a high recharge characteristic of a humid climate ($R = 381 \text{ mm/a}$ (millimeter per year)) leads to a high water table, intercepting even the highest valleys. Here, local gaining streams function as effective drains, and regional flow is suppressed. In Figure 4b, a low recharge characteristic of an arid climate ($R = 25 \text{ mm/a}$) results in a water table that is below the high valleys. Here, local flow is "switched off" and recharge is transported through regional flow toward the lower streams. The streams in the upper valleys are surface runoff fed, and if running at all, leak through their beds to the groundwater below. Unlike *Toth's* [1963] simulations, where R rate was not varied, these simulations demonstrate the effect of recharge on the water table position. Thus, it is likely that a humid climate favors local flow, and an arid climate favors regional flow.

[21] 2. The geological media must also influence the flow regime. In Figure 4c, the hydraulic conductivity is increased by tenfold from Figure 4a (0.5 to 5 m/d), a small difference given the 13 orders-of-magnitude range in natural geologic media. Although recharge remains the same, the faster drainage leads to a lower water table, switching off local flow beneath the upper valleys and transporting recharge into the lower valleys. In Figure 4d, an aquifer thickening in the downdip direction, commonly observed in sedimentary formations, further enhances regional flow, simply due to the increased flow cross section to accommodate a larger volume.

[22] 3. Underlying both *Toth's* [1963] and our own modeling studies is the fundamental assumption that the subsurface drainage boundary coincides with the surface drainage boundary at the largest scale. That is, no subsurface inflow or outflow is permitted at the model boundary, forcing all recharge to emerge as river flow at the lowest basin outlet. However, this may not be the case in a real world setting. Figure 5 illustrates a hypothetical but common situation where a surface drainage system has developed over a dipping sedimentary rock sequence. The dipping beds in general, and the carbonate rock unit in particular, can divert groundwater away from the basin, leading to complications in the theoretical concepts discussed above. First, the elevation dependence no longer holds; a major exporting basin may exist in the lower reaches (brown, over the carbonates). Second, the scale dependence is no longer clear; one observes greater groundwater export from the basin situated at midelevation, rather than the expected transition from exporting to importing moving down drainage. Note that these complications also exist where the dip direction is consistent with surface drainage but the dip angle is steeper, such as found in coastal plain settings. Third, the regional flow hinge line cannot be defined, because the surface drainage is in the opposite direction of groundwater drainage (toward the left, along bedding).

[23] Thus, when the surface and the subsurface drainage systems are incongruent, the theoretical relationships outlined above disintegrate at certain scales. The cause of this incongruity is the geological structure underlying a river network, which controls the bulk of groundwater movement, but is only partially reflected in the surface drainage development. In the subsequent analyses, this assertion must be kept in mind when interpreting the magnitude and spatial distribution of groundwater exporters and importers based on observations of the natural world.

3. Methods and Hypotheses

[24] Consider the steady state water balance of an arbitrary river basin,

$$P - ET = R = Q_r + Q_g \quad (1)$$

where P is precipitation, ET is evapotranspiration, R is basin recharge, Q_r is river outflow, and Q_g is groundwater outflow. That is, the atmospheric surplus ($P - ET$) provides input to the river basin (R) which is then partitioned into river outflow (Q_r , surface runoff plus base flow), and lateral groundwater outflow or inflow (Q_g). A simple indicator of

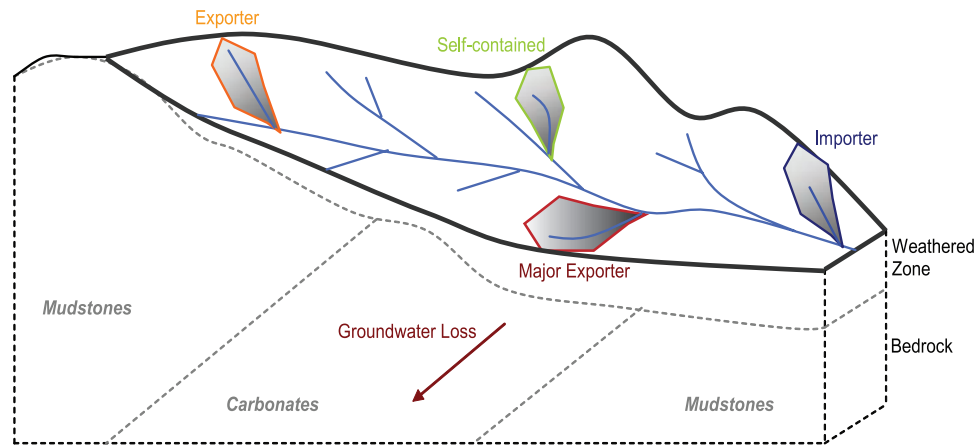


Figure 5. A hypothetical situation where the carbonate unit diverts water away from the river basin under study. In this case, the surface drainage does not coincide with the subsurface drainage, with regard to flow boundaries as well as flow directions. Lower reach basins can be exporters, complicating the “elevation dependence,” and the large basin as a whole is not self-contained, complicating the “scale dependence.”

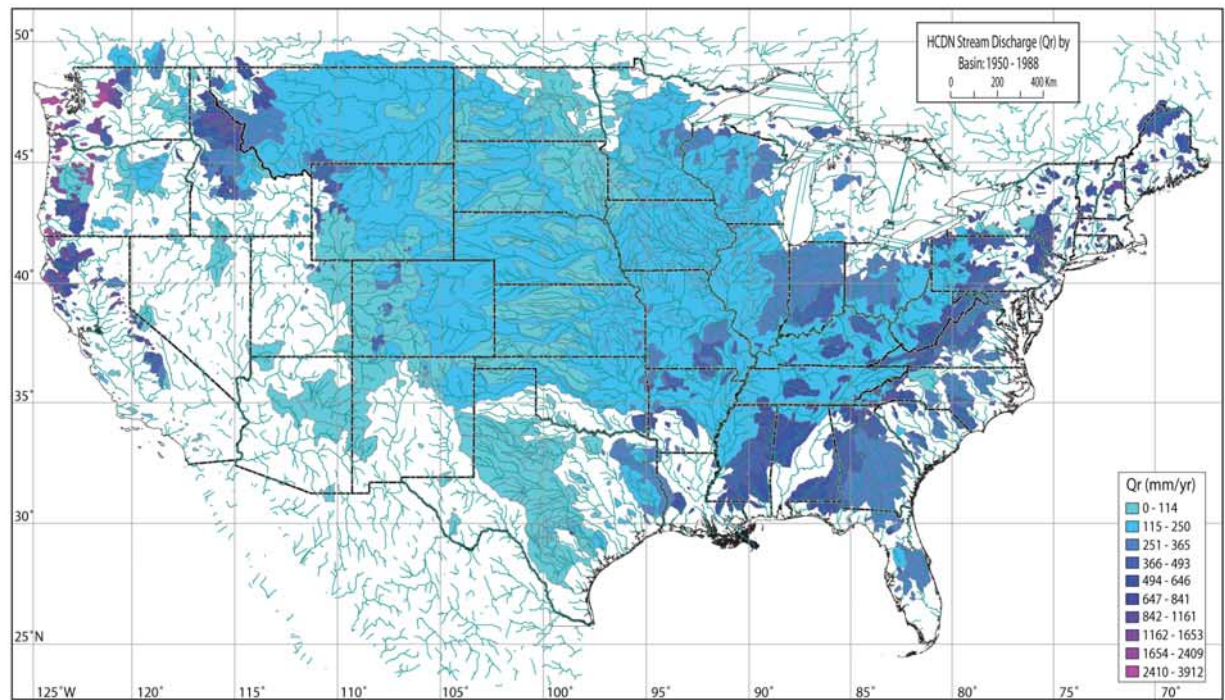
this partition is the ratio $Q_r:R$, the fraction of basin recharge (R) that exits via surface outflow. It follows that if $Q_r:R < 1$, then a portion of R failed to emerge as river outflow, and the basin must be a groundwater exporter. If $Q_r:R > 1$, the observed river flow must include groundwater inflow from other basins, and the basin is a groundwater importer. If $Q_r:R = 1$, then the imports and exports exactly balance over this basin.

[25] In the ensuing analysis, Q_r , the streamflow from a river basin, is obtained from the U.S. Geological Survey (USGS) Hydro-Climatic Data Network (HCDN) for the contiguous United States [Slack *et al.*, 1993]. Each river flow record in the HCDN data set is based on USGS observed stream discharge corrected for reservoir effects over the record period of water years 1874–1988. Figure 6a gives the location, size (3.5–2,954,000 km²), and basin mean annual streamflow (mm/a) from 1555 river basins in the HCDN database used in our analyses. We note that the observed streamflow contains the effect of groundwater pumping, which can be significant over regions where groundwater sustains base flow and is heavily mined for irrigation [see Bartolino and Cunningham, 2003, available at <http://pubs.usgs.gov/fs/fs-103-03/#pdf>]. An example is the High Plains Aquifer System (Figure 2), the largest unconfined aquifer system in the United States, where large-scale pumping for irrigation has reduced streamflow up to 30% in parts of the region [Healy *et al.*, 2007]. This reduces Q_r and the $Q_r:R$ ratio, causing erroneous attribution of groundwater pumpage to groundwater export. However, quantifying streamflow depletion over the nation is difficult because groundwater pumping for irrigation, operated by individual farmers, is poorly documented. We have begun a systematic effort to assess the link between groundwater pumping and streamflow variability at regional scales through detailed analysis of paired water table and streamflow observations. Without a more quantitative knowledge of this link at the present, we must keep this potential source of large uncertainty in mind while interpreting the results. In particular we must avoid detailed interpretation of the

$Q_r:R$ ratio in regions with documented streamflow depletion due to pumping.

[26] The basin recharge, $R = P - ET$, is obtained from observed P gridded over North America and model-calculated ET from a 50-year (1950–2000) retrospective simulation of surface water and energy balance over North America, using the Variable Infiltration Capacity (VIC) hydrology model, at a grid resolution of 0.125° longitude and latitude. Both the forcing data and the model validation and output are described in detail by Maurer *et al.* [2002]. A map of $R = P - ET$ over the VIC model domain can be found in Figure 6b. Whereas P can be observed, ET must be estimated because of the lack of large-scale observations. VIC simulated ET is based on energy and water budget calculations in the vegetation and soil, driven by observed atmospheric conditions (precipitation, air humidity and temperature, wind, and short- and long-wave radiation). Without observations, a direct validation of modeled ET is through the water budget closure so that $P - ET$ balances the stream flow. The VIC water budget compares favorably with observed river flow over a 10-year period, at monthly steps, from 12 river basins across the diverse climate-geologic conditions of the continent [Maurer *et al.*, 2002], which gives us confidence that VIC ET is reasonable. We also compared the VIC simulated groundwater recharge with several USGS estimates. Using streamflow separation, Wolock [2003] obtained a nationally consistent recharge estimate. While VIC and Wolock’s assessment agree well over the western half of the United States, VIC recharge is higher in the east and the southeast, particularly in the humid Atlantic coastal plains. Independent local estimates, also by the USGS [e.g., Martin, 1998; Delin and Risser, 2007], are closer to VIC than Wolock [2003] in the coastal plains of New Jersey (Wolock, 200–400 mm; VIC, 300–500 mm; Martin, 350–500 mm) and North Carolina (Wolock, 50–800 mm; VIC, 300–1200 mm; Delin and Risser, 20–1410 mm). Thus, VIC appears to be the only estimate that has a national coverage, is based on a nationally consistent method, and agrees well with the available USGS studies in different regions of the continent. We recognize the uncertainties associated with

(a) Observed Basin and Annual Mean River Flow (mm)



(b) VIC-Estimated Annual Mean P-ET (mm)

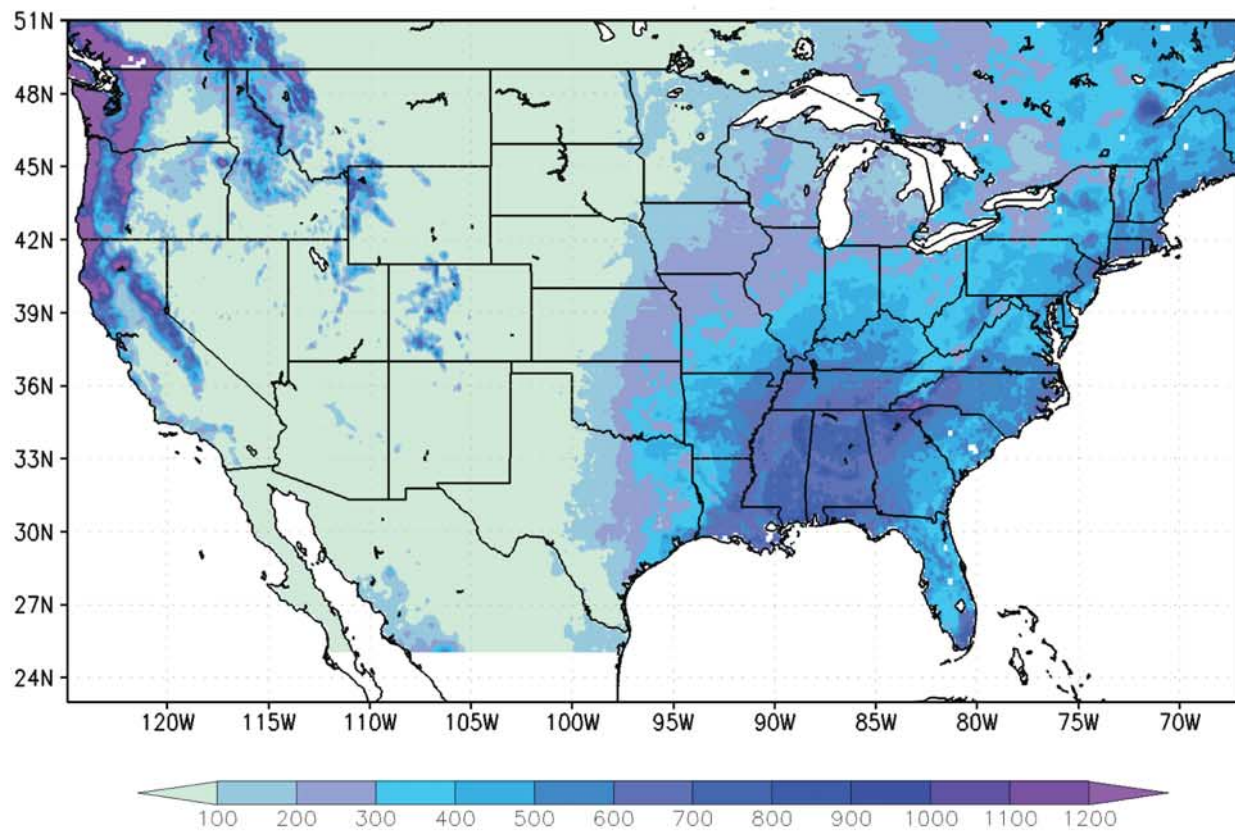


Figure 6. (a) Location of the 1555 Hydro-Climatic Data Network (HCDN) basins with basin mean annual flow (mm) of the 1555 river basins in the USGS HCDN over water years 1874–1988, and (b) VIC-estimated annual mean basin recharge $R = P - ET$ (mm), 1950–2000.

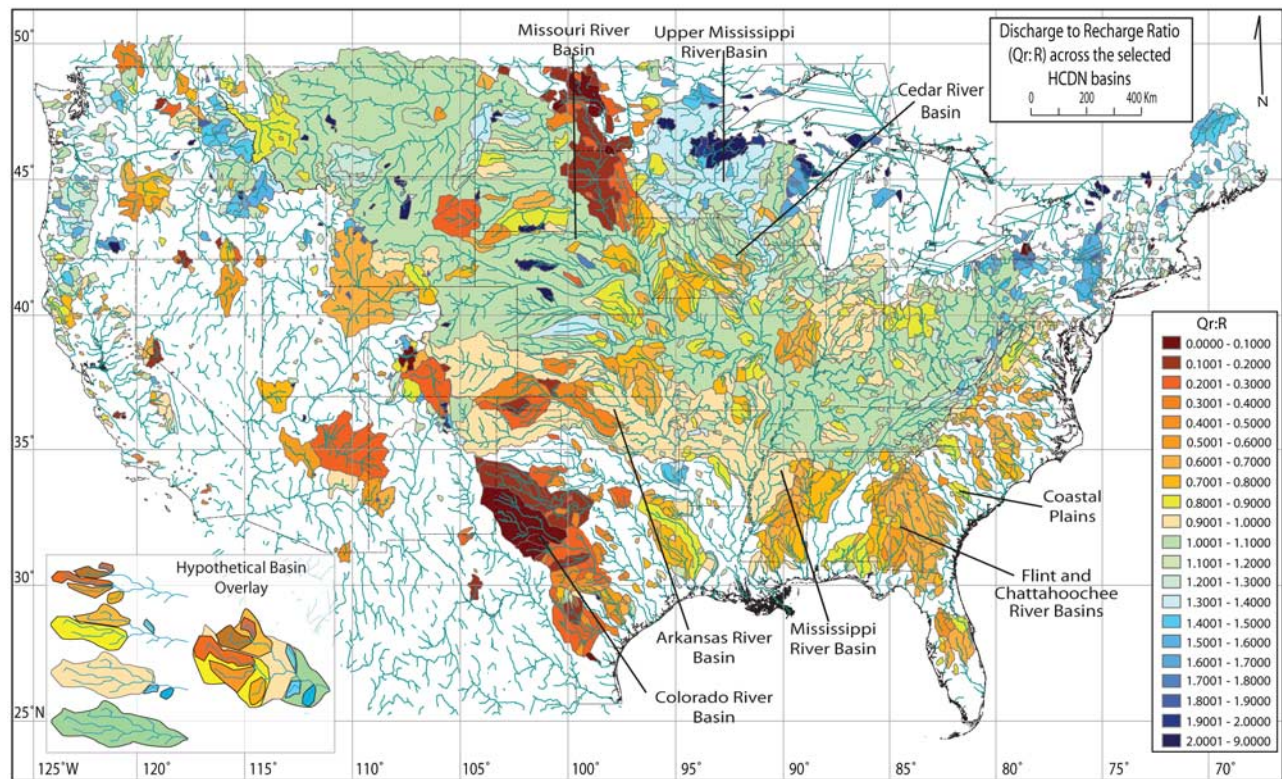


Figure 7. Map of calculated $Qr:R$ ratio for the 1555 river basins in the HCDN data set. The inset illustrates how nested basins are presented in this map, where smaller basins are overlaid atop larger basins such that a set of nested basins appears in map view as the hypothetical basin to the right.

VIC recharge estimates, but lacking better alternatives, we will use VIC for the meantime, in an attempt to obtain a first-order understanding of the significance of regional flow at the continental scale.

[27] The analysis is limited temporally to the overlap between VIC and Hydro-Climatic Data Network (HCDN) (1950–1988), and spatially to the 1555 HCDN gauging stations. The nested HCDN drainage basins (Figure 6a) were used to mask and spatially average the R values over each individual basin. The $Qr:R$ ratio is then obtained for each year over the 1555 basins and averaged over the record period, which we interpret as a measure of the climatologic mean water budget for each basin. Using this ratio, we test the following hypotheses:

[28] 1. The $Qr:R$ ratio can significantly deviate from 1, that is, groundwater export or import can be a significant part of a basin's water budget, and can be detected on the continental scale.

[29] 2. The degree of deviation of $Qr:R$ from 1 will depend on the climate; arid basins will result in larger deviations, or more significant groundwater export or import.

[30] 3. The degree of deviation of $Qr:R$ from 1 will depend on the size of the basin; small basins will exhibit larger deviations, or more significant groundwater export or import. This hypothesis has been postulated by *Genereux et al.* [2005] and here we test it at continental scales.

[31] 4. The degree of deviation of $Qr:R$ from 1 will depend on aquifer properties; thick, more permeable, and

dipping aquifers lead to larger deviations, or larger groundwater export or import.

[32] 5. The degree of deviation of $Qr:R$ from 1 will depend on the location or elevation of a basin within the larger drainage system; higher basins are likely groundwater exporters and lower basins are likely importers. For a given basin, there exists a hinge line, above which internal basins are net exporters, and below which internal basins are net importers.

[33] 6. Geologic control on the boundary of groundwater basins and the direction of flow can lead to incongruity between the surface and subsurface drainage systems. That is, the river basin and the groundwater basin can be entirely different entities. Evidence for this incongruity may provide an explanation for the observed spatial distribution in exporting and importing basins that fail to obey the theoretical relationships discussed above.

4. Results and Analyses

4.1. General Patterns and Contributing Factors

[34] The spatial distribution of the annual $Qr:R$ ratio is given in Figure 7 for the contiguous United States. The warm colored basins are where $Qr:R < 1$, indicating groundwater exporters, and the different shades of blue indicate groundwater importers. A few broad observations can be made, corresponding to the hypotheses proposed above.

[35] 1. The $Qr:R$ ratio can deviate significantly from 1 over the continent. The ratio over the basins in dark red is

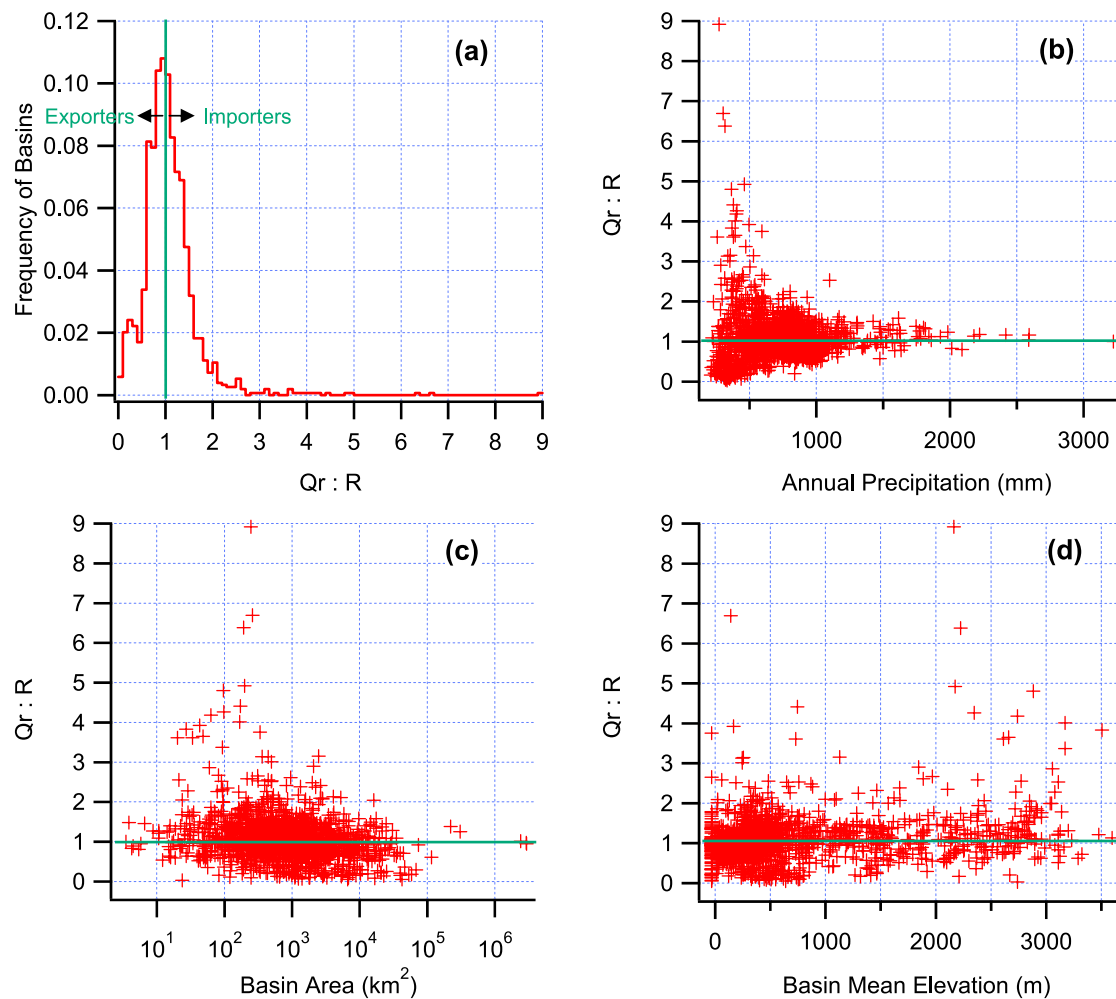


Figure 8. (a) The frequency of basins found at a given Qr:R ratio, (b) the ratio as a function of annual precipitation, (c) the ratio as a function of basin area, and (d) the ratio as a function of mean basin elevation.

<0.1 (10% of the R exits as river flow and 90% as groundwater export); the ratio over basins in dark blue can be >2 (50% of the river flow is groundwater imported from other basins). Figure 8a shows the frequency of the 1555 basins found at a given ratio. The Qr:R ratio ranges from 0.03 to 8.92, with a mean of 1.06. Half of the basins (49.6%) lie below 1 (groundwater exporters) and the other half (50.4%) above 1 (importers). However, the absolute value of the ratio for a particular basin need to be viewed with caution because of the uncertainties in both Qr and R as discussed earlier. Nevertheless, the range and the distribution of the ratios support the hypothesis that groundwater outflow or inflow can be a significant portion of a basin's water budget at the continental scale.

[36] 2. Overall, the distribution of the Qr:R ratio reflects the general distribution of climate zones with more exporting basins in the arid southwest. Figure 8b plots the Qr:R ratio against basin mean annual precipitation. The largest deviations from 1 occur at the lower end of the range, while the basin with the highest precipitation lies on 1. Thus, these results support the hypothesis that groundwater export or import is a larger portion of the water budget in more arid regions.

[37] 3. There is a tendency of the Qr:R ratio toward 1 with increasing basin area. The Qr:R ratio is plotted against basin area in Figure 8c, which shows that the largest deviations occur at basin sizes near 100 km². Figure 3 suggests that basin area and Qr:R ratio should be highly correlated; however, there is a lack of relationship here, which is due to several factors. First, the smaller basins are not nested inside larger ones, as in Figure 3 where scale dependence is expected to hold; instead, lumped together are many separate river systems spanning a wide range of climate regimes, which has a stronger influence on the ratio at the continental scale. Second, the data set includes a wide range of geologic conditions, which also influence the ratio at a given basin size. These assertions highlight the need to examine selected river basins with more uniform climate and geology, and with small basins nested within larger ones as in Figure 3.

[38] 4. It appears that thick and dipping sand-gravel aquifers tend to yield exporting basins. For example, most of the basins on the Atlantic and Gulf of Mexico coastal plains are groundwater exporters (Figure 7), where thick, highly permeable, and regionally dipping aquifers underlie and parallel the river drainage. Thus, these results support the hypothesis that efficient aquifers favor regional flow or

groundwater exporting/importing basins. This geological effect imparts secondary variability to the overall climatically controlled distribution of importing and exporting basins.

[39] 5. Within certain basins, higher elevation subbasins are exporters while lower elevation basins are importers. For example, within the larger Arkansas drainage (tan color, Figure 7), a cluster of higher-elevation subbasins (dark red to orange) are exporters, and a small cluster of subbasins in the lower reach (light green) are importers, with a hinge line lying somewhere between them. Another example is found in the Colorado basin of Texas, where strong exporters (dark red) in the upper reaches transition to moderate exporters (orange), then to weak importers (light green) near the coast. A comparison of Qr:R ratio against the basin mean elevation (Figure 8d) for the entire data set reveals no clear pattern. The reasons are that, first, the mean elevation of a large river basin is not meaningful when compared to small subbasins nested within it. Second, independent drainage systems across the continent are lumped together; although the elevation dependence may hold within a group of nested basins (Figure 3), it is unlikely to hold across different basins with different climate and geologic regimes. This again highlights the need to examine selected river basins with hierarchical nesting as in Figure 3.

[40] From Figure 8, it appears that the only relationship that holds well at the continental scale is the dependence on climate (Figure 8b). The large range of Qr:R ratio at low precipitation shows a distinct convergence toward 1 with increasing precipitation. While this relationship manifests itself clearly at the continual scale, characterized by large climate variability, it may not be so apparent at smaller basin scales where climate may change little, while the geology and topography can change greatly. This “scale dependence of climate control” will become more apparent in the subsequent analysis of individual basins.

[41] 6. The incongruity between the surface and subsurface drainage systems may be another reason for the lack of a clear scale and elevation dependence discussed in hypotheses 3 and 5. Groundwater flow independent of river drainage alters the spatial distribution of exporters and importers. An example is the Cedar River of Iowa, where the importers are situated above the exporters. To understand such cases, and to further test hypotheses 3 and 5, the scale and elevation dependence of Qr:R ratio, we undertake a more detailed study of individual basins which have well-constrained climate and elevation patterns. The following three basins are selected for this purpose because their geologic settings have been fairly well documented in the literature.

4.2. A Detailed Study of Select Basins

4.2.1. Texas Interior to Coastal River Basins

[42] Figure 9a contains a map of the Qr:R ratio over Texas. Basins enclosed by bold outlines, and those within, are examined closely. This set of basins is selected for the following reasons: First, several small basins are nested inside the larger ones as in Figure 3, facilitating comparisons with theoretical relationships. Second, these basins span the entire drainage spectrum from interior headwaters to coastal watersheds. Third, there is sufficient geologic information in the literature to infer any potential geologic

control. Fourth, the general direction of groundwater flow (Figure 9b) is southeast and toward the Gulf of Mexico [Barker *et al.*, 1994; Barker and Ardis, 1996; Grubb, 1998, 2001; Williamson and Grubb, 2001; Ryder and Ardis, 2002], largely parallel to the surface drainage and minimizing the complexity due to incongruity between surface and subsurface drainage systems. Several small coastal basins further to the east are included to augment the data set at the receiving end of the drainage. The Qr:R ratio ranges from 5% in the interior to 138% near the Gulf coast.

[43] Since there is a strong west–east precipitation gradient (Figure 9c), it is expected that climate may play a role in the spatial distribution of the Qr:R ratio. Figure 10a plots the ratio versus mean annual precipitation, which exhibits a strong correlation, indicating the importance of groundwater export in drier basins. However, the observed relationship has been strengthened by the fact that the basins on the humid end of the climate spectrum also happen to be at the lower end of the drainage and receive regional groundwater flow (triangles in Figure 10a). The effect of climate alone would be a tendency for Qr:R to approach 1 with increasing humidity, but not to exceed 1. The Qr:R ratios above 1 at these basins must be attributed to their location at the receiving end of regional flow. In this case, the west–east climate gradient is aligned with the west–east elevation gradient, both responsible for the increasing Qr:R ratio eastward toward the coast.

[44] The relationship is weakened by a cluster of basins which, although situated at the drier end of the climate spectrum (green-tan basins in Figure 9a, near 30° latitude), have much higher Qr:R ratios than expected, and some are even groundwater importers (circles in Figure 10a). The geologic cross section (Figure 9d) reveals that they are situated over the Balcones fault zone. Here, the highly permeable carbonate rocks of the Edwards-Trinity aquifer system, underlying the upper Colorado drainage, are truncated by faults, forcing groundwater to disperse upward to the land surface [Barker and Ardis, 1996]. Numerous springs exist along the Balcones fault zone, often where faulting has placed highly permeable units adjacent to impermeable units, and the general topographic relief is high [Barker *et al.*, 1994]. This upward groundwater flow leads to an abundance of small streams. Surface water exits these small basins through channels flowing over the Texas coastal uplands aquifer system, characterized by southeastward dipping and progressively thickening siliciclastic sand and silt beds [Grubb, 1998], which are once again groundwater exporters (dark red-tan basins below, Figure 9a). In this case, the combined climate and elevation gradient from west to the east is punctuated by a geological singularity, which significantly alters groundwater flow paths. The correlation coefficient of $r = 0.9233$ is obtained from data points excluding basins that lie on the fault zone.

[45] Figure 10b shows a plot of the Qr:R ratio versus basin area. In the theoretical scenario of Figure 3, the ratio should approach 1 with increasing area, but this is not observed here. We see a negative correlation where larger basins exhibit larger groundwater loss. This unexpected pattern may exist because the largest basins also happen to be in the driest climate and at the highest elevations (Figure 9a), both leading to low Qr:R ratios. Another key cause may be the lack of smaller basins nested within larger

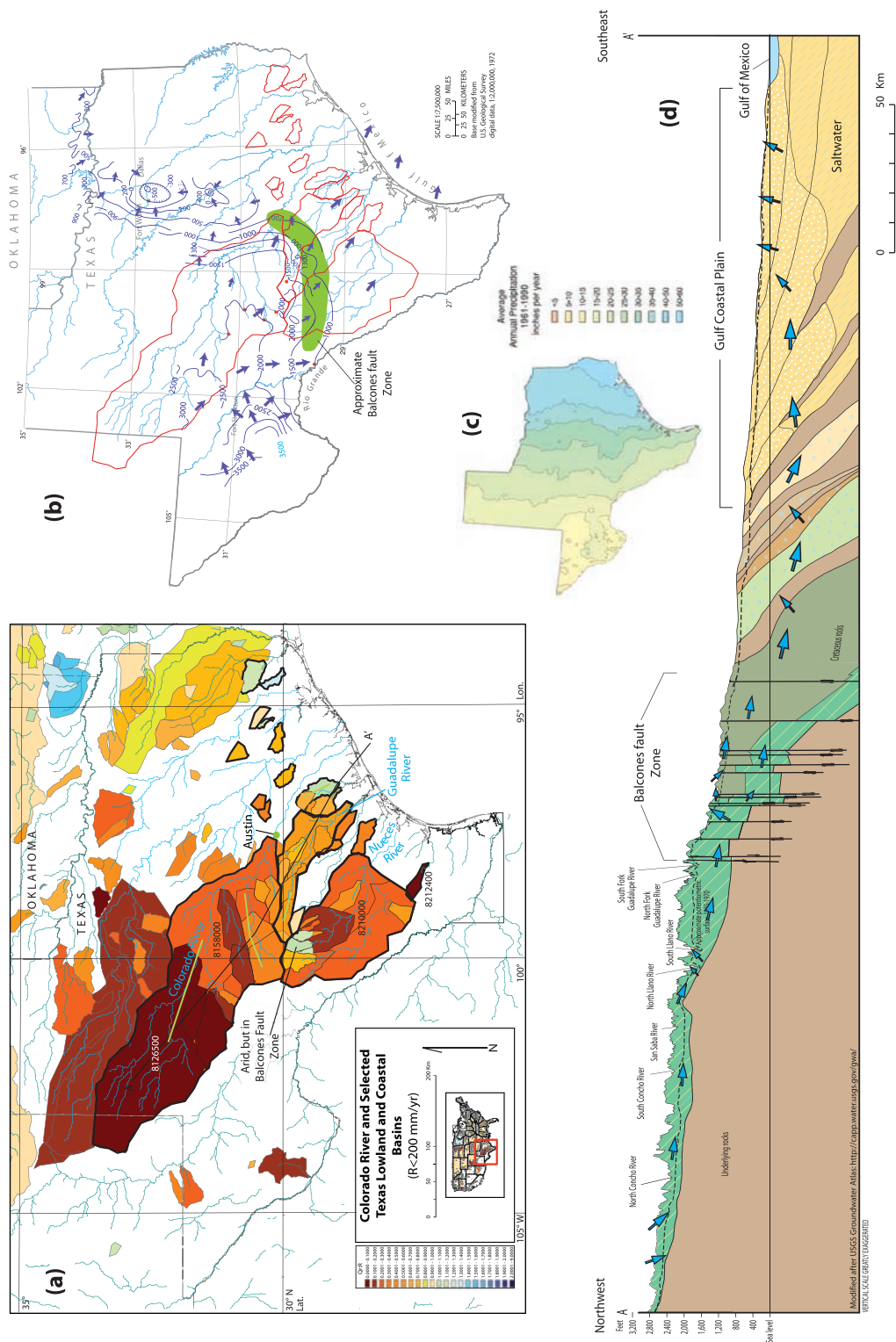


Figure 9. (a) The Q_r/R ratio over Colorado River and lowland and coastal basins in Texas, (b) equipotential lines and direction of groundwater flow, (c) annual precipitation (from <http://nationalatlas.gov/mld/prism0p.html>), and (d) a geologic cross section.

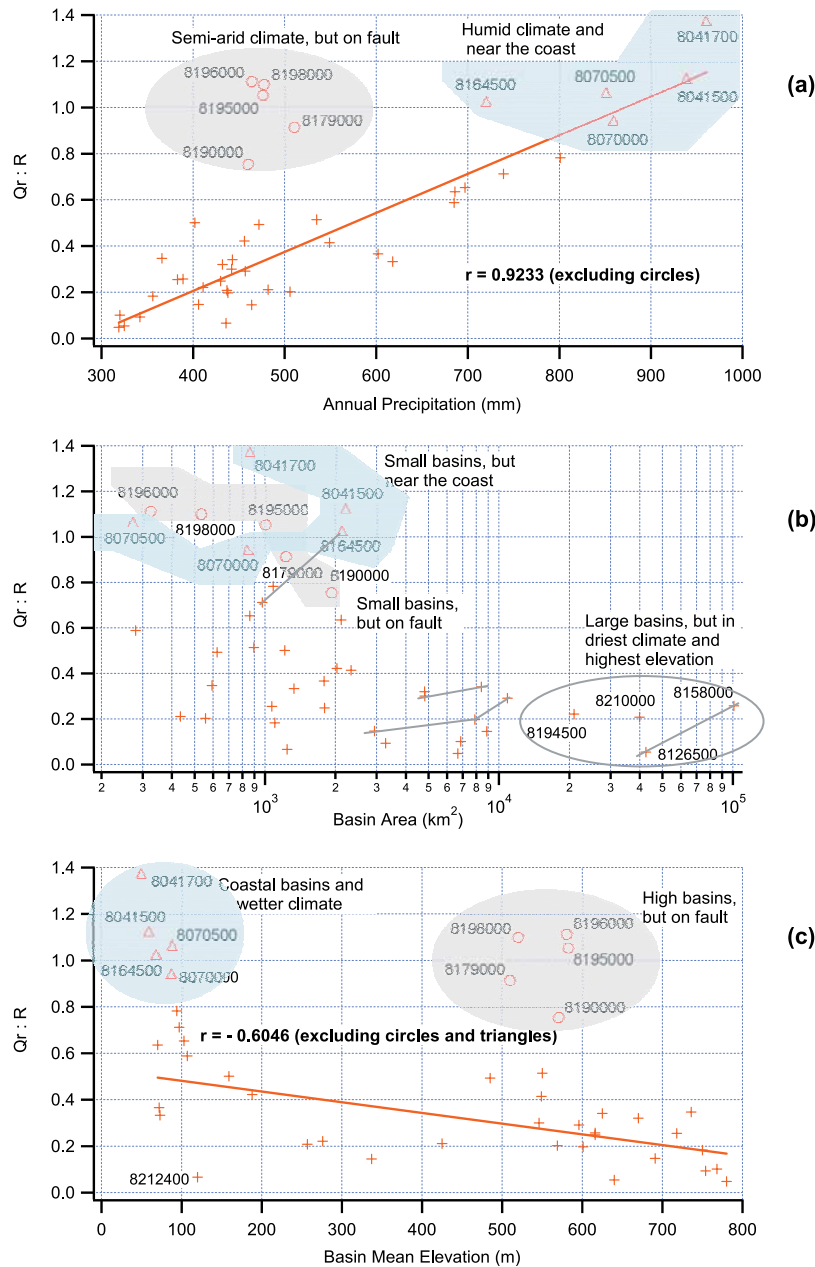


Figure 10. Relationship of Qr:R with (a) climate, (b) basin area, and (c) basin elevation for the Texas basins.

ones, as in the theoretical case of Figure 3. If the basins are strictly nested, then there is indeed a tendency of Qr:R toward 1, as shown by the gray lines in Figure 10b and the green lines on the map in Figure 9a, both connecting the few pairs of basins satisfying the strict nesting condition. Otherwise, the largest basins do not fully encapsulate all the basins analyzed, and thus do not include the down-gradient portion of the flow system. Here, the strong climate and elevation influence obscures the scale effect across the region, but the relationship is locally pervasive where strict nesting exists under similar climate-elevation conditions.

[46] Comparison of Qr:R ratio versus basin mean elevation in Figure 10c shows a significant negative correlation, as theory predicts. This relationship is strengthened by a group of coastal basins (which also happen to receive

more precipitation) with Qr:R ratios near 1 (triangles, Figure 10c), but it is weakened by a cluster of high-elevation basins over the Balcones fault zone where groundwater emerges to feed rivers (circles, Figure 10c). The latter are minor exporters and some even importers, elevating the Qr:R ratio toward 1. Also weakening the relationship is a small basin in the coastal plain (8212400, the southernmost basin in Figure 9a) which is a significant groundwater exporter. The likely cause is the high permeability of coastal plain aquifers which, combined with the arid climate and the relatively high elevation (120 m above sea level, higher than all other coastal basins), produces a low Qr:R ratio. Indeed, it has been suggested that the upper coastal basins recharge the underlying aquifer and those close to the coast receive discharge from the regional

flow system [Williamson and Grubb, 2001]. Thus, in this case the elevation gradient is aligned with the climate gradient in controlling the Qr:R ratio, which is reversed locally by the geologic control of the Balcones Fault Zone and the efficient subsurface drainage of the coastal plain aquifers. The correlation coefficient of $r = -0.6046$ is obtained without the humid coastal and the fault-aligned basins (pluses only).

[47] We may define a hinge line separating the groundwater recharge zones of the interior and upper coastal Texas from the discharge zones of the lower coastal plain. Unfortunately, insufficient numbers of HCDN basins within close proximity of the coast prevent us from determining if this is a region of groundwater discharge. However, as can be seen in Figure 9a, a zone of abrupt increase in drainage density along the gulf coast is readily evident. Slightly updip is likely the hinge line transition separating groundwater exporting basins from importing basins.

4.2.2. Flint and Chattahoochee River Basin, Georgia, Alabama, and Florida

[48] Figure 11a shows a map of the Qr:R ratio over the Flint and Chattahoochee River basins (outlined in bold). The basins were chosen for a detailed study for three reasons. First, the smaller basins are nested inside the larger ones. Second, the direction of groundwater drainage is aligned with the surface drainage system toward the Gulf of Mexico. Third, although the geologic setting is similar to that of the Texas basins discussed above, the climate is much more humid (twice as much precipitation), which may help in determining the role of climate in positioning the regional hinge line. The Qr:R ratio ranged from 46% to 112% over these basins.

[49] The Qr:R ratio is plotted against annual precipitation in Figure 12a. Theory predicts that as precipitation increases, the ratio should converge toward 1. The lack of such a relationship here may be attributed to three factors. First, the basins with the highest rainfall (triangles, Figure 12a) also happen to be at the highest elevations; Figure 12b plots basin elevation versus precipitation, which suggests a strong orographic influence on local precipitation. This trend breaks near the coast where precipitation increases again because of sea breeze effects (basin 2357000). While high rainfall leads to a Qr:R ratio approaching 1, the high elevation places these basins at the exporting end of the drainage cascade, reducing the ratio to below 1. In this case, contrary to the Texas basins, the climate gradient works against the elevation gradient in controlling the Qr:R ratio, and as a result, the expected climate dependence is weakened by the elevation dependence, and vice versa. The relatively weak overall precipitation gradient (~from 840 to 1000 mm/a) may also influence this trend. The second factor explaining the lack of relationship in Figure 12a is the group of two small basins on the fall line (Figure 11a), the boundary between continental bedrock and coastal plain sediments. These two basins have unusually high Qr:R ratios (circles, Figure 12a), likely caused by the thin soil and high topographic relief in the bedrock terrain above the fall line, restricting groundwater flow to local systems. The third factor is the group of two basins with unusually low Qr:R ratios (squares, Figure 12a). Figure 11b and the geologic cross section show that the Flint-Chattahoochee River basin terminates over the highly permeable rocks of

Floridian carbonate platform, the surface outcrop of which appears as a “U” shape in the southernmost portion of the basin. The Floridian Aquifer is riddled with karst giving groundwater a matrix of pipelines through the subsurface; it has been identified as a major conduit of regional flow from the interior coastal plain (as well as Florida) to the Gulf [Miller, 1986; Barker and Pernik, 1994]. In this case, the climate influence is reversed by the elevation influence, and further obscured by the abrupt geologic transition across the fall line in the middle reach and the great capacity of the Floridian Aquifer to carry groundwater flow at the lower reach.

[50] Figure 12c plots the Qr:R ratio against basin area, which lacks the expected dependence on scale: a convergence toward 1 with increasing basin size. The same three factors can be evoked to explain the lack of a clear relationship. Figure 12d shows the Qr:R ratio plotted against basin mean elevation. The lack of the predicted decrease in Qr:R with increasing elevation, contrary to theory, is again attributable to the same three factors.

[51] Compared to the Texas basins, the Flint and Chattahoochee River basins differ in three important ways. First, in the Texas basins, the climate and topographic gradient influence the Qr:R ratio in the same direction, reinforcing one another and producing a strong climate and elevation dependence. In Flint and Chattahoochee the two factors interfere destructively with one another, such that the net effect is an apparent lack of the expected relationships. Second, the average climate over the Texas basins is much drier than Flint and Chattahoochee (≈ 500 mm versus 1000 mm of annual precipitation, respectively), and as a result, the overall Qr:R ratio is much lower (49% versus 70%), despite the rather similar geologic setting (bedrock aquifers transitioning into coastal plain sediments). Third, the more humid climate of the Flint and Chattahoochee basin might also influence the position of the regional hinge line. Because of the humid climate and high basin recharge, the regional groundwater flow system is able to reach volumes great enough to force the saltwater-freshwater interface into the Gulf, facilitating direct submarine groundwater discharge off the coast [Renken, 1996]. Radon isotope tracer studies have confirmed significant submarine groundwater discharge in the Gulf coast region, initiated by both the highly permeable Floridian Aquifer, and the thick, unconsolidated sands of the Coastal Plain [Cable *et al.*, 1996]. Anomalously high fluxes of phosphorus and nitrogen nutrients offshore in the Gulf of Mexico have been attributed to submarine groundwater discharge, increasing productivity in areas of locally high influx [Slomp and Van Cappellen, 2004]. Thus, it seems likely that the Flint-Chattahoochee River basin as a whole sits above the regional hinge line, and the highly permeable seaward dipping geology coupled with increased recharge has effectively shifted the hinge line offshore of the continent and pushed the salt wedge seaward.

4.2.3. Cedar River Basin, Iowa

[52] The Cedar River flows through northeastern Iowa and into the upper Mississippi River (Figure 13a). It was selected for a closer study because of the counterintuitive arrangement of the internal basins: the upper basins are groundwater importers (shades of green) and the lower basins are exporters (orange-tan colored), opposite of the

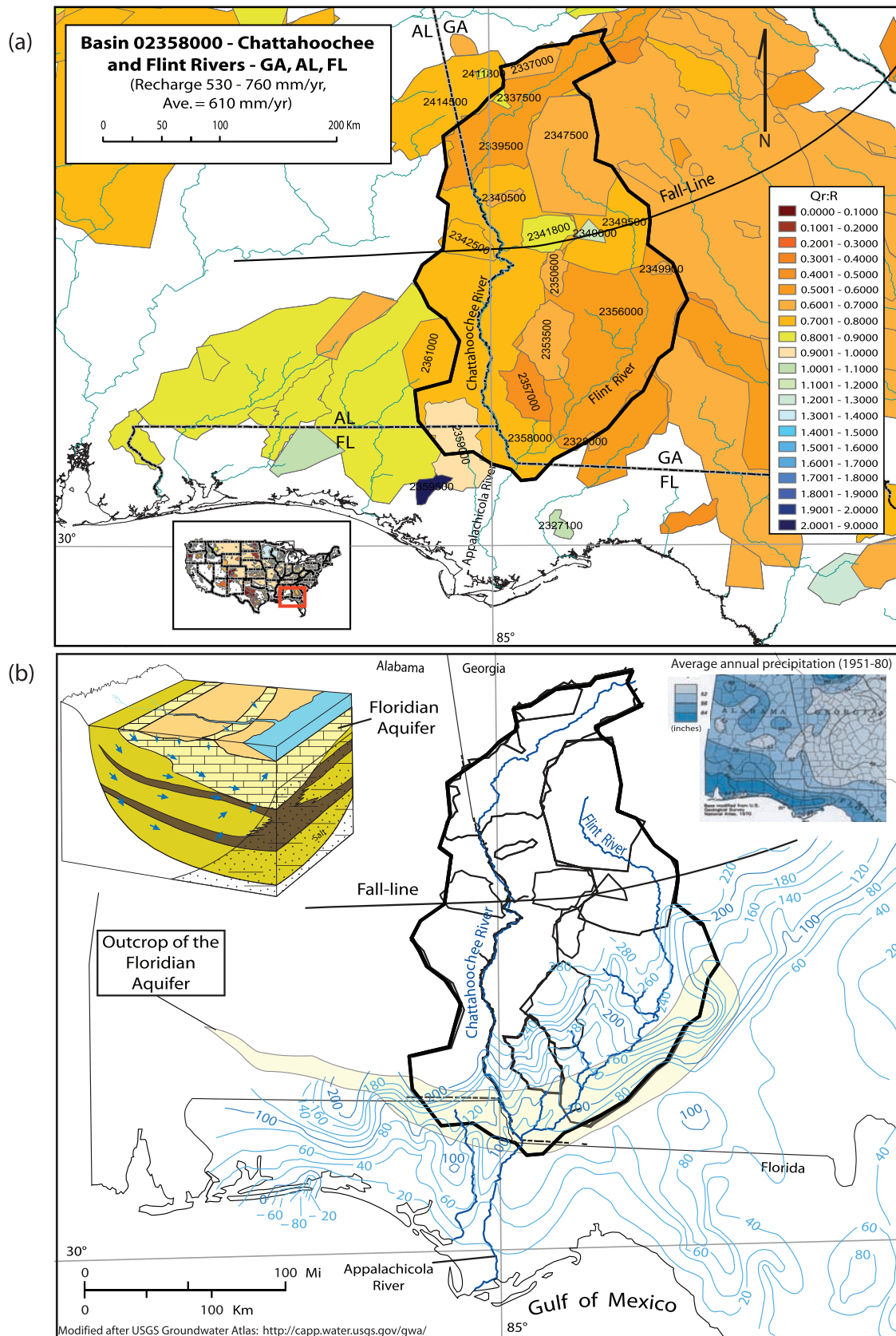


Figure 11. (a) The Qt:R ratio over Flint and Chattahoochee River basins, and (b) groundwater equipotential lines with a geologic block diagram and precipitation map shown as insets.

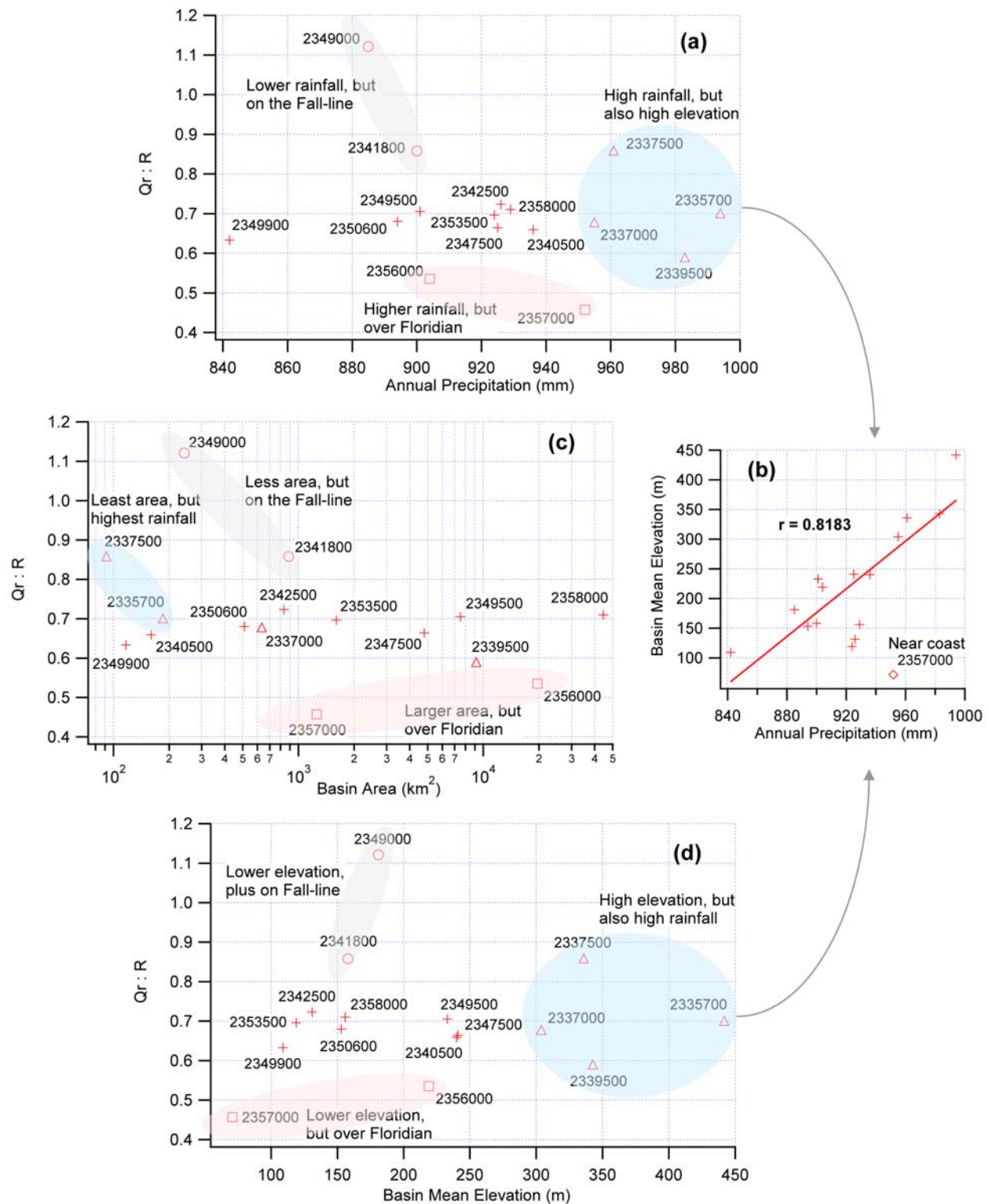


Figure 12. (a) Relationship of Qr:R with climate, (c) basin area, and (d) basin elevation, and (b) the correlation between precipitation and elevation in the basin for the Flint and Chattahoochee River basins.

expected spatial relationship. The Qr:R ratio ranged from 60% in the lower basins and 138% in the headwaters.

[53] The cause of the reversal lies in the region's complex geology [Young and Siegel, 1992], and in particular, a

sequence of highly permeable carbonate rocks underlying different sections of the basin (Figure 13b). The importing headwater basins of the Cedar River sit over the outcropping Devonian carbonate rocks (dark blue in Figure 13b),

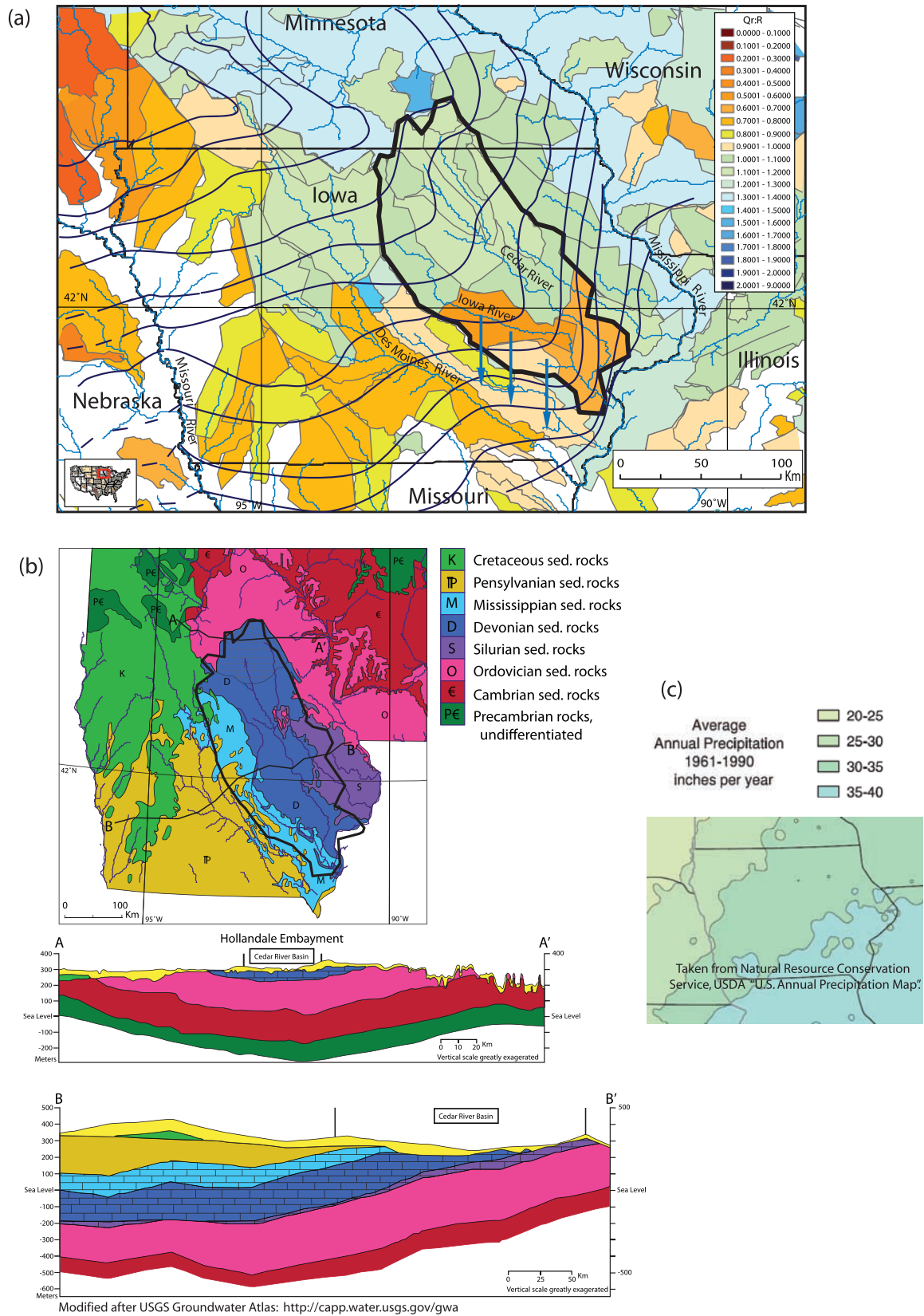


Figure 13. (a) The Qr:R ratio and groundwater equipotential lines, (b) geologic map and cross sections, and (c) precipitation gradient over Cedar River basin.

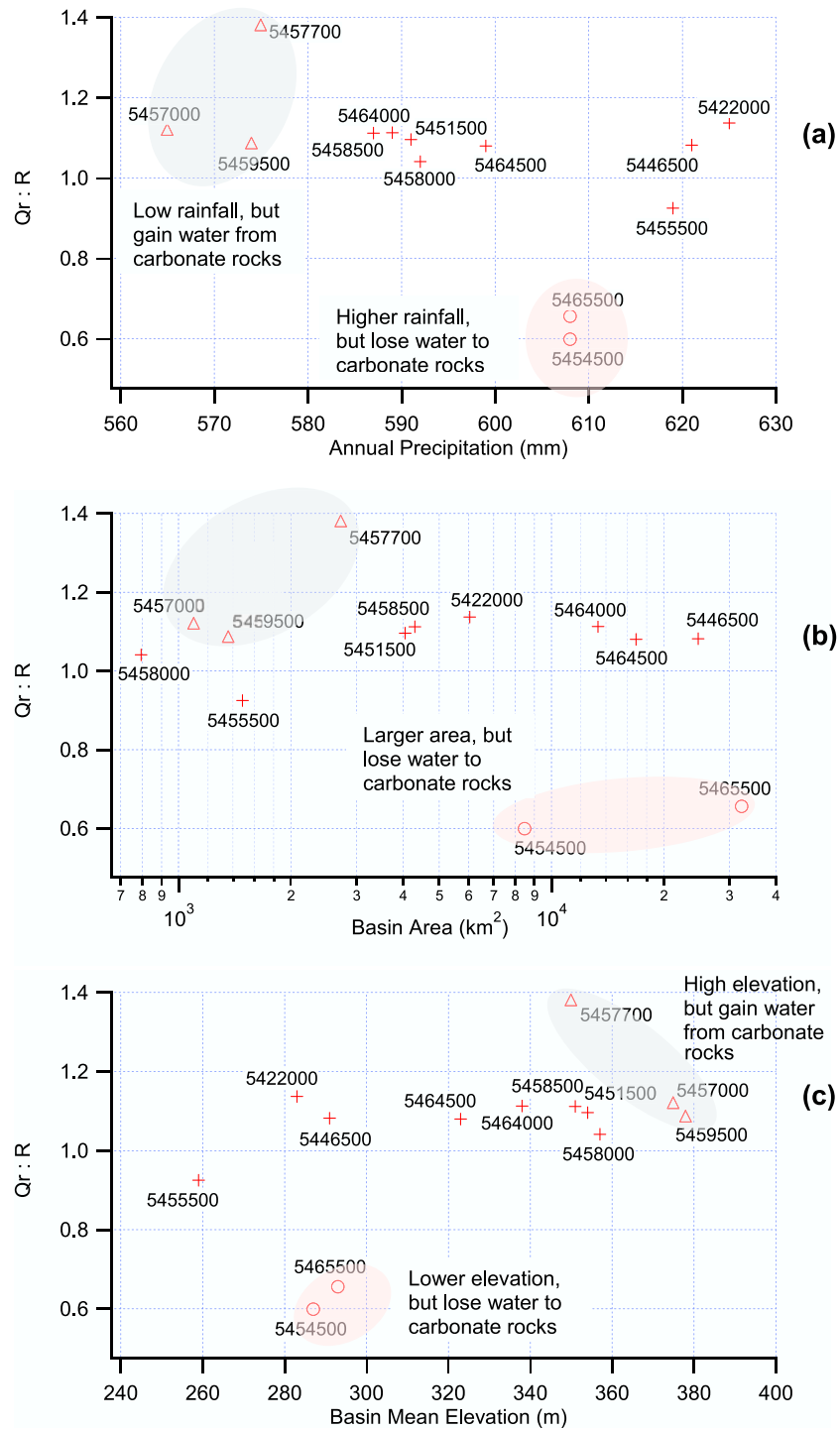


Figure 14. (a) Relationship of $Qr:R$ with climate, (b) basin area, and (c) basin elevation over Cedar River basin.

the uppermost unit of the “U” shaped basin strata shown in line A–A’, known as the Hollandale Embayment. This bowl-shaped formation extends beyond the topographic divide and acts as a collection pan, receiving recharge over a larger area than the surface catchment of the basin. The formation is underlain by the less permeable Ordovician sedimentary rocks, thus forcing groundwater to emerge at the center of the bowl and spring feed the headwater

channels of Cedar River, giving the latter unusually high flow volume.

[54] As shown in section B–B’, midway through the basin, the Devonian carbonate stratum thickens and dips toward the southwest. Onlapping the Devonian stratum are the highly permeable Mississippian carbonate rocks (light blue, Figure 13b) underlying the lower half of the Iowa River basins (joining Cedar River further southeast). These thickening and dipping carbonate strata provide an efficient

groundwater conduit toward the south, diverting flow in the subsurface and giving rise to the large groundwater exports in the southern half of the basin.

[55] Figure 14a shows the $Qr:R$ ratio plotted versus basin annual precipitation. With a west–east precipitation gradient (Figure 13c), the ratio should be below 1 in regions of low rainfall and approach 1 with increasing rainfall, as in the Texas basins (where the elevation gradient is aligned with the rainfall gradient). However, the strong geologic control in the Cedar River Basin seems to override the joint climate–elevation influence; the expected pattern is reversed due entirely to geology. Figure 14b shows the $Qr:R$ ratio versus basin area. Again, elevated ratios in the headwaters (triangles, Figure 14b), and lowered ratios over the lower basins (circles, Figure 14b) obscure the expected relationship. Figure 14c gives the ratio versus basin mean elevation. Despite the west–east climate gradient that should strengthen the elevation influence, the pattern is reversed.

[56] Groundwater flow under the Cedar River is complex and largely reflects the underlying geology. The broader picture given by the potentiometric contours in Figure 13a suggests that the Cedar River is only part of a larger regional flow system, flowing northwest to southeast, with a source area roughly in the Dakotas [Mandle and Kontis, 1992]. These data provide further evidence that geology may define the boundaries of recharge and discharge. Here, the direction and amount of groundwater flow often has little to do with the surface drainage system, which largely follows the land surface topography. Thus, the groundwater basin and surface water basin are not synonymous. In cases such as this, geologic factors must be invoked to understand the pattern of groundwater flow and its role in a river basin's water budget.

5. Summary and Discussion

[57] The focus of this paper is long-distance groundwater flow, which transports water from one river basin to another and discharges to rivers and wetlands far distant from the recharge area. We asked two questions: How much water exits a basin without ever passing through the surface outlet of that basin? Under what climatologic and geologic conditions is this flow component significant? Our objective was to assess the significance of this flow term in affecting the water budget of a river basin, and the factors that control its magnitude. We approached our objective through two stages. First, with conceptual model simulations, we extended *Toth's* [1963] theories of multiple-scale groundwater flow, and the role of topography and basin depth, to include the influence of climate and geology. Second, by using the $Qr:R$ ratio, where Qr is observed river flow and R is model-simulated basin recharge, we calculated the magnitude of groundwater export ($Qr:R < 1$) and import ($Qr:R > 1$) over the lower 48 states of the United States. Three basins are selected for a detailed study to elucidate the role of climate, scale, elevation, and geology in controlling exporting and importing. Here we summarize our preliminary findings.

[58] First, within the uncertainties in both Qr and R in the method used here, regional groundwater flow can be a significant part of a basin's water budget. The $Qr:R$ ratio ranged from <0.1 , or 90% of incipient water exiting via

regional groundwater flow, to >2 , or 50% of river flow originating from other basins via regional groundwater flow.

[59] Second, climate exerts a clear and predictable influence on the importance of regional groundwater flow, particularly on the continental scale. Large groundwater exports and imports occur in arid regions, where the water table normally lies below the local topography in the upper drainage, and emerges at the valleys in the lower drainage. This results in large upland regions with deep groundwater, and smaller lowland patches with wetlands and abundant streams. Overall, the distribution of importing and exporting basins align quite well with climatic trends on the continental scale (Figure 8b), with secondary complications due to geology. This leads to the generality that climate (through recharge) may be used as a first-order predictor of $Qr:R$ distributions on these broad scales.

[60] Third, the size of the basin exhibits a certain degree of influence on the importance of groundwater export or import; indeed, imports and exports nearly balance in the largest continental-scale basins where direct submarine groundwater discharge is absent (such as the Mississippi river basin). However, its role is obscured in our data set because basins of different sizes are not nested (not all smaller basins drain into larger ones) and span regions with large climatic and geologic variability. Thus, although one can expect a larger basin to be more self-contained, the climatic and geologic changes often occur at smaller spatial scales and override the signal of scale influence.

[61] Fourth, the relative position or elevation of a basin in the drainage continuum, namely, headwaters versus coastal zones, has a strong influence on the relative amount of groundwater import and export. For example, in the Texas basins where geologic control is limited to a small area in the fault zone, the higher elevation interior basins are exporters and the lower elevation coastal basins are importers, although this elevation influence is accentuated by a climatic gradient.

[62] Fifth, the geology underlying a basin has emerged as the most important regional-scale control over the $Qr:R$ distribution. On the continental scale, the secondary effects of geology impart variability to the primarily climate induced spatial distribution of exporting-importing basins, such as over the Atlantic and Gulf coastal plains (Figure 7). On the regional to basin-scale, geology is responsible for the numerous and effective outliers in the theoretical relationships (Figures 10, 12, and 14). These outliers, such as geologic disruptions (e.g., faulting, the fall line), highly efficient subsurface flow conduits (e.g., carbonate rocks, coastal plain sand gravel), or incongruity between surface network and groundwater basins, weaken and sometimes entirely reverse the expected smooth transitions from exporting to importing basins across climate, scale, and elevation gradients. At these smaller scales, climatic effects add secondary variability to the overall geologic control.

[63] Our findings suggest that pathway 6 (Figure 1), regional groundwater flow, can be a significant part of a river basin's water budget under certain climatic and geologic conditions across the continent. The existence of regional or interbasin groundwater flow, its significance in sustaining regional stream base flow, and its role in supporting arid and coastal wetlands, are well known to hydrologists working at watershed and small basin scales. However,

these effects are less well known to the large-scale water cycle and climate modeling communities, and their significance has not been assessed at the continental scale. Our preliminary work presented here is intended to highlight the potential importance of this process and the need for further investigation.

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